

Know Your Blind Spots: Better Visibility Through EDR Policy Hardening

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Abstract

Endpoint Detection and Response (EDR) tools identify, detect, and respond to anomalous behavior. They assist blue teams, incident response operations, and threat hunting. However, an EDR is only as effective as the events it can detect. Alerts and actions depend on the tool's detections, which in turn depend on visibility within the environment. Analysts can identify only threats visible to them, which depends on properly configured tools. Modern EDRs use configurable policies to determine which logs and telemetry are collected. These policies often start with default settings that minimize environmental impact. Vendors recommend adjusting configurations to match the organization's security policy. But what value do these settings provide beyond their descriptions? By using the Atomic Red Team library, the unique data captured by each policy can be listed and mapped to the MITRE ATT&CK Framework. This approach provides defenders and threat hunters with clearer insights into the visibility potential of each setting.

1. Introduction

Last year, I was part of a team of analysts conducting a threat hunt at a large public organization. Our efforts focused on scrutinizing their EDR logs for known indicators of compromise (IOC), anomalous activity, and potential behavioral indicators based on threat intelligence. At the end of the hunt, we identified only a few low-priority findings and a couple of recommendations to improve their cyber posture. Two months later, another team discovered PRC-based BRICKSTORM malware within the environment we had just hunted and determined that the Threat Actor (TA) had been present for over a year. Our team had not detected their activity—not a single indicator or alert to suggest the network had been compromised. The question we were left asking ourselves was, “How could this have been possible?”

A threat analyst can only be as good as the data they have available to analyze. Following this hunt, the team reviewed our tools and determined that the organization had not configured its EDR security policies beyond the default, out-of-the-box settings. These default settings resulted in visibility gaps, leading to missed detections. The descriptions for these policies provide a general idea of what additional features they include. However, they do not offer specific details that defenders and threat hunters can plan against. If the organization's security team had a clearer understanding of the value of these settings, they might have been more inclined to implement them. Enabling some of these policies would have enhanced their detection and alerting capabilities, allowing them to identify the adversary's presence sooner and potentially avoid the incident entirely. Knowing what additional data and network visibility each of these policy settings provides can help decision makers determine how best to implement those configurations.

EDR policies can affect user notifications when an alert is triggered, detection capabilities based on available telemetry, and even the level of preventive measures taken

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to protect the system when malicious activity is detected. This enhanced security comes with a cost, though, as prevention methods can impact system-level performance and increase the risk of false-positive detections, leading to denial-of-service on legitimate activity. Resource requirements notwithstanding, a properly configured and optimally tuned EDR can be an incredibly powerful tool for threat analysts to detect adversary activity. By measuring data-collection deltas between default and hardened configurations using adversary emulation and mapping the results to the MITRE ATT&CK Framework, to what extent can we increase visibility and detection and improve Threat Hunting effectiveness?

1.1. Purpose

To understand the value of individual security policies in our EDR, it is necessary to evaluate how each setting affects visibility. Changes in visibility resulting from incremental policy changes can be measured by comparing observed detections and field names/values. These visibility deltas can then be mapped to the MITRE ATT&CK Framework, allowing for a clearer understanding of the value each policy offers analysts and how it relates to adversary activity. "As a detection engineer, you should be able to predict the data that you get to work with, as well as understand exactly what is available - especially before you get started with unintentional blind spots" (Bluenomicon., 2023, p. 54).

1.2. Scope

This effort focuses on the CrowdStrike Falcon EDR and measures its visibility into a Windows 11 endpoint. CrowdStrike Falcon leverages a robust security policy rollout with three implementation phases, each phase enabling additional security policies on protected endpoints within the environment. This phased approach makes it the preferred test case for measuring security policies applied to a standalone Windows 11 endpoint, leveraging the Atomic Red Team library of adversary simulations. Most security features on this host were disabled to maximize EDR detection capabilities.

2. Research Methodology

2.1. EDR Discussion

As the name suggests, EDR tools secure endpoints using installed agents. These agents parse event data in real time to detect indicators of compromise (IOCs) and behavioral patterns, and prevent harmful or malicious applications from executing if configured to do so. When a detection occurs, the EDR tool generates an alert, and designated security personnel receive notifications to review and take follow-on action if deemed necessary. This EDR activity, along with detection alerts and endpoint telemetry, is aggregated and stored in accordance with the EDR retention policy. All this available data can then be queried and reviewed by security analysts looking for unexpected or suspicious activity that may have gone undetected by the EDR.

An alternative description of viewable data is 'visibility.' For this research, visibility will be defined as the event fields available for querying under the implemented security policy, and the data within those fields. This visibility is what threat hunters and security hunters use to determine how to search for adversaries within the environment.

To determine what to monitor for detections, what activities to prevent, and what notifications to send, EDRs leverage comprehensive security policies. Each policy enables certain features – additional notifications and alerting, increased detection capabilities, enhanced or more strict real-time prevention settings. Many of these features also add additional data points to the telemetry gathered from each endpoint. All this endpoint data is normalized into a database with pre-labeled fields containing metadata, properties, and sub-properties. The data can then be manipulated and presented to the user in the form of dashboards, reports, or other visualizations, or queried based on specific criteria.

2.1.1. EDR Security Policies

CrowdStrike offers three pre-configured security policies that they recommend implementing in a phased approach. This phased approach allows organizations time to

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evaluate each policy for potential environmental impacts and to tune policies based on their needs. [Appendix 1](#) describes CrowdStrike implementation recommendations for these policies.

For this research, these pre-configured policies offer a convenient way to measure changes in visibility over a cross-section of security settings.

2.2. Atomic Red Team

The Atomic Red Team project is an open-source library of tests designed to emulate common adversary behavior. Each test is short and focuses on a single (sub-)technique from the ATT&CK framework, and according to the project website, “each test runs in five minutes or less.”

The Atomic Red Team library includes 328 unique tests, which are all tied to the MITRE ATT&CK Framework. During testing, CrowdStrike security settings prevented two of the tests from running. These were documented, and testing continued after removing the offending test from the atomic library.

```
Exit code:
Done executing test: T1003.001-1 Dump LSASS.exe Memory using Procdump
Executing test: T1003.001-2 Dump LSASS.exe Memory using comsvcs.dll
Exception calling "Start" with "0" argument(s): "Access is denied"
At C:\Users\joshu\OneDrive\Documents\WindowsPowerShell\Modules\Invoke-AtomicRedTeam\2.3.0\Private\Invoke-Process.ps1:45 char:17
+ $process.Start() > $null
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : Win32Exception

Exit code:
Done executing test: T1003.001-2 Dump LSASS.exe Memory using comsvcs.dll
Executing test: T1003.001-3 Dump LSASS.exe Memory using direct system calls and API unhooking
The system cannot find the path specified.
Exit code: 1
Done executing test: T1003.001-3 Dump LSASS.exe Memory using direct system calls and API unhooking
Executing test: T1003.001-4 Dump LSASS.exe Memory using NanoDump
The system cannot find the path specified.
Exit code: 1
Done executing test: T1003.001-4 Dump LSASS.exe Memory using NanoDump
Executing test: T1003.001-6 Offline Credential Theft With Mimikatz
Exception calling "Start" with "0" argument(s): "Access is denied"
At C:\Users\joshu\OneDrive\Documents\WindowsPowerShell\Modules\Invoke-AtomicRedTeam\2.3.0\Private\Invoke-Process.ps1:45 char:17
+ $process.Start() > $null
+ ~~~~~
+ CategoryInfo          : NotSpecified: (:) [], MethodInvocationException
+ FullyQualifiedErrorId : Win32Exception

Exit code:
Done executing test: T1003.001-6 Offline Credential Theft With Mimikatz
Executing test: T1003.001-7 LSASS read with pypykatz
The system cannot find the path specified.
Exit code: 1
Done executing test: T1003.001-7 LSASS read with pypykatz
Executing test: T1003.001-8 Dump LSASS.exe Memory using Out-Minidump.ps1
```

Figure 1: Atomic Red Team Running in PowerShell

2.3. MITRE ATT&CK Framework

The ATT&CK Framework by MITRE is a matrix of tactics, techniques, and procedures (TTPs) that adversaries use to conduct their operations. Separate matrices exist for enterprise, mobile, and Industrial Control Systems (ICS) environments. The Enterprise framework consists of 216 techniques and 475 sub-techniques across 14 tactic groups. Each technique and sub-technique includes an overview of the methodology, with recommendations for both a detection strategy and mitigation controls.

EDR detections are frequently mapped to the ATT&CK Framework when presented to analysts. As new visibility data points are identified, they can also be mapped to the ATT&CK Framework to provide additional value. This mapping can be completed manually or by correlating the EDR event timestamp with the Atomic log file timestamp on the testing VM.

2.4. Environment Design

A free trial of CrowdStrike Falcon was used to complete this research. The trial allowed access to their online portal and to download the Falcon agent installation file. The default trial configuration was used, which includes Falcon Prevent and Falcon Device Control. Additional features such as Falcon Insight XDR, Falcon FileVantage, or Falcon Discover were not requested or available. These would have added additional insights into visibility, but are outside the scope of this research.

A Dell Latitude 7430 with an Intel Core i7 CPU and 32 GB of RAM was utilized. Testing was completed on a single Virtual Machine (VM) with the recommended settings for running Windows 11 25H2 Professional. The OS was configured for personal use and included a virtual TPM. As part of the initial operating system setup, all Windows privacy settings (Location, Find my Device, Diagnostic Data, Inking & Typing, and Tailored Experiences) were turned off, and VMware Tools were installed for functionality. A personal Microsoft account was used for simplicity and to potentially capture cloud-based information. Lastly, Microsoft update KB5077241 was installed, and no other updates were installed during the research. Sufficient resources were not

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available to implement a full domain environment, which could have generated additional telemetry.

Several prerequisites were necessary for the Atomic Red Team to run properly. The Atomic library was installed in C:\atomic-red-team-master, .NET Framework 3.5 was installed, and Windows Defender was turned off (remember, the goal is to run as many tests as possible without interruption so that the Falcon EDR can capture related events). The following script was used to run all the atomic library tests unattended.

```
Invoke-AtomicTest All -PathToAtomicsFolder "C:\atomic-red-team-master\atomics" -Force -ErrorAction SilentlyContinue
```

The script directs PowerShell to run every test in the Atomic library located in the specified Folder Path, forces the command to override restrictions, suppresses error messages, and prevents the script from failing when encountering errors. A folder named C:\Exclusions was created, and the script was saved there under a file named "Invoke-Atomic".ps1. CrowdStrike was configured to recursively ignore the "C:\Exclusions" folder, forcing the EDR to allow the script running from that folder to automate test execution in the atomic library. PowerShell's script execution policy was set to unrestricted to ensure Windows would also allow the script to run.

```
Set-ExecutionPolicy Unrestricted
```

It was necessary to disable Microsoft OneDrive to prevent Atomic tests from interacting with personal files and causing errors that would affect the tests.

```
Stop-Process -Name "OneDrive" -Force -ErrorAction  
SilentlyContinue
```

2.5. Research Design

Using a comparative research approach will allow identification of changes in visibility across iterations of Atomic Red Team libraries. A baseline will be established with only notification settings enabled and all other security policies disabled. The testing

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VM will then be reverted to the baseline snapshot, and simulations will be run against each successive security policy. Visibility will be measured by querying for all fields that contain data observed during the current test run, and repeating this for each iteration. The Falcon EDR can be queried for these values using the `fieldset()` and `fieldstats()` commands. Different event types can also be enumerated by querying the `#event_simpleName` field.

CrowdStrike documentation includes a 'data dictionary' that specifies what events or logs each field captures. Additionally, evaluating sample values from the unique fields identified in this visibility research will provide a general sense of what each field captures. CrowdStrike security policies fall under several categories: Sensor Capabilities, Sensor Visibility, Next-gen Antivirus, Malware Protection, and Behavior-based Prevention. CrowdStrike also uses a sliding scale of sensor sensitivity for each of these categories, including: Off (Disabled), Cautious, Moderate, Aggressive, and Extra Aggressive. Each tier purportedly incurs additional performance impacts, which should be weighed against the benefits of the added detection capabilities before implementation.

2.6. Limitations

Different EDR tools implement security policies in different ways, and the results herein will be specific to CrowdStrike Falcon. Other EDR solutions will likely require additional testing. The testing environment also included a single Windows 11 VM, as previously mentioned. Visibility likely varies depending on the environmental variables available across different networks and configurations, such as Active Directory or Windows Server features. The testing was done across several iterations of varying durations, possibly because the system was not dedicated and other processes were running concurrently.

The Atomic Red Team includes a comprehensive library of tests, but it does not reflect all adversary activities. Observed results will confirm a correlation between the visibility gained from a given security policy. Still, a lack of results does not provide

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adequate confidence that security policies offer additional visibility, simply because no delta is observed. The Atomic tests may lack the required prerequisite activities to generate the correct log or trigger the appropriate detection(s).

3. Findings and Discussion

3.1. Observations and Results

Early on in this research, scope creep would be a significant consideration. CrowdStrike security policies have additional sensitivity settings for each security category, as described previously. Consequently, there is an exponential number of policy combinations that cannot be fully enumerated within the constraints of this research. Therefore, efforts were focused on the policies contained in the sensor visibility category. Testing was done against each security policy, first with sensitivity disabled, and again with sensitivity set to the maximum value, 'extra aggressive.' This was done to more accurately determine which fields and events fall under each policy setting, given the appropriate sensitivity level.

Tests were also run against the three pre-configured policy groups recommended by the vendor for phased implementation to provide additional data comparisons. Referring to the phased policy groups also makes it easier to make recommendations to organizations, as it provides a clearer understanding of the benefits of each policy.

Full disclosure: the results were not consistent across multiple test iterations. There were only a few cases where the same fields were observed under the same policy across all research. Results are organized by policy and include vendor documentation descriptions.

3.1.1. Additional User Mode Data Visibility

Description: “Allows the sensor to get more data from a user-mode component it loads into all eligible processes, which augments online machine learning and turns on additional detections.”

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Additional User Mode Data (AUMD) was far and away the most notable policy setting that gained visibility indicators. Enabling this policy added 102 unique event fields to our EDR data, compared to the baseline. A majority of these (75) were different sub-properties of the assessments field, which hold Boolean values.

Field Name	Field Type	Example Value
assessments.additional_user_mode_data_enabled	Boolean	"yes" "no"
assessments.automated_remediation	Boolean	"yes" "no"
assessments.beta_build_disabled	Boolean	"yes" "no"
assessments.branch_target_injection_mitigation	Boolean	"yes" "no"
assessments.branch_target_injection_mitigation_hardware_support	Boolean	"yes" "no"
assessments.branch_target_injection_mitigation_patch	Boolean	"yes" "no"
assessments.branch_target_injection_mitigation_registry_allowed	Boolean	"yes" "no"
assessments.credential_guard_running	Boolean	"yes" "no"
assessments.debug_mode_disabled	Boolean	"yes" "no"
assessments.dma_guard_enabled	Boolean	"yes" "no"
assessments.engine_full_visibility	Boolean	"yes" "no"
assessments.execution_blocking_custom_blocking_enabled	Boolean	"yes" "no"
assessments.execution_blocking_intel_threats_enabled	Boolean	"yes" "no"
assessments.execution_blocking_suspicious_processes_enabled	Boolean	"yes" "no"
assessments.execution_blocking_suspicious_registry_ops_enabled	Boolean	"yes" "no"
assessments.execution_blocking_suspicious_scripts_enabled	Boolean	"yes" "no"
assessments.exploit_mitigation_force_aslr_enabled	Boolean	"yes" "no"

assessments.exploit_mitigation_heap_spray_allocation_enabled	Boolean	"yes" "no"
assessments.exploit_mitigation_null_page_allocation_enabled	Boolean	"yes" "no"
assessments.exploit_mitigation_seh_overwrite_protection_enabled	Boolean	"yes" "no"
assessments.exploitation_behavior_application_exploitation_activity_enabled	Boolean	"yes" "no"
assessments.exploitation_behavior_chopper_webshell_enabled	Boolean	"yes" "no"
assessments.exploitation_behavior_code_injection_enabled	Boolean	"yes" "no"
assessments.exploitation_behavior_driveby_download_enabled	Boolean	"yes" "no"
assessments.exploitation_behavior_javascript_execution_rundll32_enabled	Boolean	"yes" "no"
assessments.firmware_is_uefi	Boolean	"yes" "no"
assessments.hardware_enhanced_exploit_detection	Boolean	"yes" "no"
assessments.hsti_available	Boolean	"yes" "no"
assessments.http_detections	Boolean	"yes" "no"
assessments.hvci_enabled	Boolean	"yes" "no"
assessments.hvci_strict_mode	Boolean	"yes" "no"
assessments.in_full_functionality	Boolean	"yes" "no"
assessments.interpreter_only	Boolean	"yes" "no"
assessments.iommu_available	Boolean	"yes" "no"
assessments.iommu_in_use	Boolean	"yes" "no"
assessments.kmci_enabled	Boolean	"yes" "no"
assessments.l1_terminal_fault_mitigation	Boolean	"yes" "no"
assessments.lateral_movement_credential_access_credential_dumping_enabled	Boolean	"yes" "no"
assessments.lateral_movement_credential_access_windows_logon_bypass_enabled	Boolean	"yes" "no"
assessments.mbec_available	Boolean	"yes" "no"
assessments.ml_adware	Boolean	"yes" "no"

assessments.ml_adware_prevention	Boolean	"yes" "no"
assessments.ml_cloud_antimalware	Boolean	"yes" "no"
assessments.ml_cloud_antimalware_prevention	Boolean	"yes" "no"
assessments.ml_sensor_antimalware	Boolean	"yes" "no"
assessments.ml_sensor_antimalware_prevention	Boolean	"yes" "no"
assessments.quarantine_and_security_registration	Boolean	"yes" "no"
assessments.quarantine_on_write	Boolean	"yes" "no"
assessments.ransomware_backup_deletion_enabled	Boolean	"yes" "no"
assessments.ransomware_cryptowall_enabled	Boolean	"yes" "no"
assessments.ransomware_file_encryption_enabled	Boolean	"yes" "no"
assessments.ransomware_file_system_access_enabled	Boolean	"yes" "no"
assessments.ransomware_locky_enabled	Boolean	"yes" "no"
assessments.real_time_response_enabled	Boolean	"yes" "no"
assessments.rogue_data_cache_load_mitigation	Boolean	"yes" "no"
assessments.rogue_data_cache_load_mitigation_patch	Boolean	"yes" "no"
assessments.script_based_execution_monitoring_enabled	Boolean	"yes" "no"
assessments.script_enforcement	Boolean	"yes" "no"
assessments.secure_boot_enabled	Boolean	"yes" "no"
assessments.secure_kernel_running	Boolean	"yes" "no"
assessments.secure_mor_available	Boolean	"yes" "no"
assessments.sensor_tampering_protection	Boolean	"yes" "no"
assessments.smm_protections	Boolean	"yes" "no"

assessments.speculative_store_bypass_mitigation_available	Boolean	"yes" "no"
assessments.speculative_store_bypass_mitigation_hardware_support	Boolean	"yes" "no"
assessments.suspicious_kernel_drivers	Boolean	"yes" "no"
assessments.system_firmware_bios_enabled	Boolean	"yes" "no"
assessments.test_signing_disabled	Boolean	"yes" "no"
assessments.uefi_memory_protection	Boolean	"yes" "no"
assessments.volume_shadow_copy_audit	Boolean	"yes" "no"
assessments.volume_shadow_copy_protect	Boolean	"yes" "no"
assessments.vsm_available	Boolean	"yes" "no"
assessments.windows_insider_program_disabled	Boolean	"yes" "no"
assessments.windows_insider_program_not_running	Boolean	"yes" "no"
assessments.windows_os_build	Boolean	"yes" "no"
device_control_applied	Boolean	"true" "false"
device_control_applied_date	string	Date:time
device_control_assigned_date	string	Date:time
device_control_policy_id	string	id
event_type	string	"ZeroTrustHostAssessment"
group_id	string	id
group_type	string	"dynamic"
host_hidden_status	string	"VISIBLE"
os_version	string	iOS 26.3.1, Windows Server 2019, Windows 11
prevention_applied	Boolean	"true"
prevention_applied_date	string	Date:time
prevention_assigned_date	string	Date:time
prevention_policy_id	string	id
product_name	string	"Windows Server 2019 Standard"
product_type_desc	string	"Server"
remote_response_applied	Boolean	"true" "false"
rtr_state	string	"enabled"

scores.modified_time	string	Date:time
scores.os	Integer	100
scores.overall	Integer	97
scores.sensor	Integer	95
scores.version	string	4.0.7
sensor_update_applied	Boolean	"true" "false"
sensor_update_applied_date	string	Date:time
sensor_update_assigned_date	string	Date:time
sensor_update_policy_id	string	id
sensor_update_policy_type	string	"sensor-update"

Table 1 - Unique AUMD Policy Values

3.1.2. Interpreter-Only Visibility

Description: “Provides visibility into malicious PowerShell interpreter usage.”

There were seven fields exposed when testing against the Interpreter-only visibility policy. All seven were properties of the ‘attributes’ field, including `attributes.action_target_name`, `attributes.actor_cid`, `attributes.success`, `attributes.target_cid`, `attributes.target_name`, `attributes.target_uuid`, and `attributes.trace_id`.

Field Name	Field Type	Example Value
Attributes.action_target_name	string	<email> <username>
Attributes.actor_cid	string	<id>
Attributes.success	string	"SUCCESS"
Attributes.target_cid	string	<id>
Attributes.target_name	string	<email> <username>
Attributes.target_uuid	string	<id>
Attributes.trace_id	string	<id>

Table 2 - Unique Interpreter-only Policy Values

These fields all seem to populate together when an `#event_simpleName` of ‘Event_AuthActivityAuditEvent’ is spawned. This event type is created when a user logs into the CrowdStrike Falcon platform, which suggests it is not unique to this policy; rather, these data points are metadata from the EDR.

3.1.3. System Management Engine

Description: “Provides visibility into malicious System Management Automation engine usage by any application. Requires Interpreter-Only.”

Unexpectedly, no new data points were revealed in the system management engine policy, despite multiple simulations run via PowerShell. It is worth noting that this policy is dependent on interpreter-only visibility also being enabled.

3.1.4. Script-based Execution Visibility

Description: “For hosts running Windows 10 and Server 2016 and later, provides visibility into suspicious scripts and VBA macros in Office documents. Requires Quarantine & Security Center Registration toggle to be enabled.”

Three unique fields were observed when this setting was enabled: AssociatedFile, IOCType, and IOCValue. These fields are determined to be used by CrowdStrike for detection alerting, as the associated #event_simpleName for the events these fields are observed in is named “Event_EppDetectionSummaryEvent”.

Field Name	Field Type	Example Value
AssociatedFile:	string	<file_path\filename.exe>
IOCType:	string	"hash_256"
IOCValue:	string	<hash value>

Table 3 - Unique Script-based Execution Visibility Policy Values

As these fields result from an EDR detection, related fields include details on relevant MITRE TTPs, including the tactic, technique, and technique ID. Being already mapped to the MITRE ATT&CK Framework means this information is quite valuable for threat hunters and security professionals alike.

3.1.5. HTTP Visibility and Detection

Description: “Allows the sensor to monitor unencrypted HTTP traffic and certain encrypted HTTPS traffic on the sensor for malicious patterns and generate detection events on non-Server systems.”

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None of the tests returned unique fields with this setting enabled. Based on the description, I attribute this to the tests that were run not being adequate to trigger the events specific to this policy. Additional simulations utilizing HTTP and browser activities would have been more likely to reveal data points unique to this policy.

3.1.6. Hardware-Enhanced Exploit Detection

Description: “Provides additional visibility into application exploits by using CPU hardware features that detect suspicious control flows.”

No unique fields were observed across any of the iterations where hardware-enhanced exploit detection was enabled. This is likely due to the test run from the Atomic library, and additional data would be enumerated under different simulations with more explicit hardware interactions.

3.1.7. Enhanced Exploitation Visibility

Description: “For hosts running Windows 10 1809 and Server 2019 and later, [enhanced exploitation visibility] provides additional visibility into common exploitation techniques used to weaken or circumvent application security.”

Another unexpected result was that no unique fields were observed across any of the iterations with enhanced exploitation visibility enabled.

3.1.8. Enhanced DLL Load Visibility

Description: “For hosts running Windows Server, [enhanced DLL load visibility] increases sensor visibility of loaded DLLs. Improves detection coverage and telemetry, but may cause a small performance impact.”

Once again, no unique fields were observed across any of the iterations with enhanced DLL load visibility enabled. This is especially curious because multiple atomic simulations performed DLL injection.

3.1.9. WSL2 Visibility

Description: “Provides visibility into WSL2 distributions by enabling a Falcon sensor plugin.”

Interestingly, 15 unique fields were identified during initial research tests and later observed under a different policy setting when configured for extra-aggressive sensitivity. I have not yet determined how to interpret these results, other than that it is an anomaly.

3.1.10. Memory Scanning with CPU

Description: “Allows Memory Scanning to use the CPU or virtual CPU when an integrated GPU is not available.”

The only new fields observed were:

`Attributes.old_policy_settings_CPUMemoryScan`, and

`Attributes.policy_settings_CPUMemoryScan`, which occur anytime a policy change is made and reflect the most recent changes made to assigned security policies – in this case, the `CPUMemoryScan` setting. Values in these fields are strings, in the format:

```
"{\<setting>\":\<changed_value>\", \<additional_setting>\":\<changed_value>\"}"
```

3.1.11. BIOS firmware deep visibility

Description: “Provides visibility into BIOS. Detects suspicious and unexpected images.”

Similar to the previous policy change, the only new fields observed were

`Attributes.old_policy_settings_FirmwareAnalysisExtraction` and

`Attributes.policy_settings_FirmwareAnalysisExtraction`, which aligns with what was previously observed.

It is worth noting that the EDR portal maintained a cumulative count of detections, which continued to increase as testing was performed. Though not the focal

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point of this research, it indicates that the tool was working and that detections were being made.

3.2. ATT&CK Mapping

Most of the unique fields identified in this research contained EDR metadata, which does not map cleanly to the MITRE ATT&CK Framework. Fields such as the `Attributes.old_policy_settings_*` could potentially be used to detect T1562.001 – Disable or Modify Tools when unauthorized policy changes are made.

The three unique fields visible with the script-based execution visibility policy already have MITRE TTPs mapped to them under other fields in a detection-based event. This also indicates that these fields are part of post-processed metadata that analysts can review but may not need to analyze in detail.

With the Interpreter-only visibility policy enabled, the seven unique events observed are determined to be related to EDR login events. As such, there is little value in mapping to MITRE. However, we can still make some low-fidelity associations, nonetheless.

Technique	ID	Tactics
Valid Accounts	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access
Brute Force	T1110	Credential Access
OS Credential Dumping	T1003	Credential Access
Multiple	TA0008	Lateral Movement

Table 4 – Interpreter-only policy mapping

Lastly, we can map the unique fields that were identified with the AUMD Visibility policy enabled. Once again, most of these fields contain EDR metadata that contributes little to hunting efforts beyond access to endpoint data.

Overall, not much useful data was obtained that can be effectively mapped to MITRE ATT&CK.

3.3. Impact & Considerations

Given the sheer volume of additional data, enabling the AUMD policy setting has the greatest impact on visibility. Script-based execution visibility and its dependent policy, interpreter-only visibility, appear to offer the most valuable fields observed, at least for threat hunting. Having pre-existing automated detections is handy, but parsing the data manually is often a more effective approach to threat hunting.

One important consideration when enabling security policies, in addition to the greater visibility, is the likelihood of increased false positive alerts. More data means more events that can trigger alerts, and without proper tuning, that can cause significant alert fatigue. Analyst fatigue and burnout can lead to additional operating costs, including the need for security personnel to review and triage alerts and analyze log files. Additional analysts will require training to be effective, along with other personnel-related costs.

Lastly, EDR vendors charge for various service levels and access to different features. Cloud-based EDR tools are Platform-as-a-Service (PaaS) offerings that offer rapid deployment, scalability, and generally all-inclusive pricing. Conversely, on-premises tools are typically licensed only, and organizations are responsible for all provisioning and resource requirements, including storage. This independence can come at a high cost, depending on the volume of logging, which is determined by the level of security policies enabled.

With on-premises deployment, organizations will want to conduct thorough policy testing to assess resource impacts, measure alert volume, and identify policies that collect sensitive data for the organization. These metrics can then be combined with visibility statistics, as outlined through this research, to determine whether the policy is worth implementing by comparing the visibility value of each policy to the resource investment overhead.

3.4. Impactful Policy Changes

The following chart (Table 5) highlights the total number of fields observed for each policy test, and the total elapsed time taken for the full Atomic Red Team libraries to complete against that policy. These metrics are an example of measuring the ROI for implementing different policies and may need to be expanded to include additional resource considerations depending on the environment.

Policy Tested	Elapsed Time	Avg Time Deviation	Fields Observed	Avg Field Deviation
Memory Scanning with CPU	3:17:00	+25:47	456	+94
HTTP Visibility Detection	2:42:00	-09:13	439	+77
Additional User Mode Data	2:45:00	-06:13	413	+51
Script-based Execution Visibility	2:54:00	+02:47	404	+42
Quarantine & Security Center Registration	2:54:00	+02:47	404	+42
BIOS Firmware Deep Visibility	3:15:00	+23:47	403	+41
Enhanced Machine Learning for Larger Files	2:53:00	+01:47	364	+2
WSL2 Visibility	2:49:00	-02:13	337	-25
Enhanced DLL Load Visibility	2:51:00	-00:13	337	-25
Hardware-enhanced Exploit Detection	2:50:00	-01:13	329	-33
Enhanced Exploitation Visibility	2:40:00	-11:13	319	-43
Interpreter-Only Visibility	2:54:00	+02:47	319	-43
Redacted HTTP Detection Details	2:33:00	-18:13	277	-85
System Management Engine Visibility	2:40:00	-11:13	273	-89
Average	2:51:13		362.43	

Table 5 - Impact of Policy Changes

From this data, it can be inferred that AUMD, HTTP visibility, Script-based Execution Visibility, and its dependent policy, Quarantine & Security Center Registration, are more highly recommended. Memory Scanning with CPU has the highest overall number of observed fields, with 456 total fields observed; the trade-off is a potentially increased impact on system resources, having taken the longest time to complete the testing (3 hours 17 minutes).

Enhanced Exploitation Visibility, WSL2 Visibility, and Enhanced DLL Load Visibility provide lower visibility gains; they are low-impact and worth implementing for the additional telemetry. Redacted HTTP Detection is included in this list; however, it

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does not provide additional visibility; rather, it is intended to sanitize HTTP fields in the EDR interface, a useful feature worth considering for implementation.

3.5. Related Work

Existing literature on EDR tools tends to focus on detection capabilities and performance, with comparatively limited attention given to how configuration choices impact visibility at a granular level. Nevertheless, some papers do discuss EDR configurations in other ways, such as “The Proof is in the Pudding: EDR Configuration Versus Ransomware” by Benjamin Powell (2024), or “Who Needs a Pentest: Validating the Configuration of an EDR Solution Using the MITRE ATT&CK Framework” by Adam Fowler (2023).

3.6. Additional Thoughts

While the data collected through this research process ultimately proved less useful than expected, the methodology is relatively sound. The utilization of a trial-period-based EDR platform had some limitations. Not all features were included, and it is unclear what effect enabling those additional features may have had on the research. Testing against a single, non-domain Windows VM also yielded limited results compared to what is observed in larger enterprise environments.

4. Recommendations and Implications for Future Research

4.1. Recommendations

It is difficult to make specific recommendations based on such low-fidelity data. Security policies should be reviewed to determine their applicability to the organization's needs. Each policy should be tested to measure its visibility, detection capabilities, and resource impacts, as this research suggests. Testing should also include validation of the detection and prevention capabilities of each policy setting. Security policies should then

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be implemented to the greatest extent possible in accordance with individual risk tolerance and organizational requirements.

4.2. Future Research

In the future, similar research should include alternative EDR platforms to understand their visibility better. Consideration should also be given to future updates, as results may differ in newer versions of those tools. Additional testing should also be performed against different operating systems, including Linux and MacOS.

Future research should also expand on adversary emulation to more thoroughly measure visibility. The testing environment could also be expanded to resemble a full enterprise environment, including domain services and different server types. Enabling additional EDR features would also generate additional data points to measure against.

5. Conclusion

Despite what the data revealed – or did not reveal, there is still a sound model for measuring visibility based on security policy settings. Datasets differed over multiple incremental policy changes, and those deltas were measurable. Given the proper testing scenario(s) and environment, adequate data would be collected that would achieve the goal of this research.

While the original thesis could not be proven (or disproven) within the scope of this research, detection capability is critical in preventing cyber incidents, and visibility within an operating environment remains vital to that effort. Consistently and repeatably measuring visibility helps security teams understand the value of security policies and how they affect visibility.

Armed with this understanding, organizations can make more informed decisions regarding which security policies to apply, and threat hunters gain increased confidence in their detections and improved effectiveness at identifying malicious activity and

responding to threats through visibility that is more clearly defined and detection capabilities that are directly mapped to that visibility.

References

- Hortung, O. (2023). Data-Driven Detection Engineering: Making Sure You Have What You Require. M. Baccio & A Streetman (Ed.), *Bluenomicon* (pp. 48-54). Splunk.
- Rodoplu, A. (April, 2023). CrowdStrike Prevention Policy's Features and Assigning Policy to the Host Group. <https://www.alirodoplu.com/crowdstrike-prevention-policys-features-and-assigning-policy-to-the-host-group>
- Powell, B. (October, 2024). *The Proof is in the Pudding: EDR Configuration Versus Ransomware* [White paper]. SANS Technology Institute. <https://sansorg.egnyte.com/dl/wYIWUigrIw>
- Fowler, A. (September, 2023). *Who Needs a Pentest: Validating the Configuration of an EDR Solution Using the MITRE ATT&CK Framework* [White paper]. SANS Technology Institute. <https://sansorg.egnyte.com/dl/GzeRcuHWuC>
- LeBlanc, E. (February, 2025). Identifying Advanced Persistent Threat Activity Through Threat-Informed Detection Engineering: Enhancing Alert Visibility in Enterprises [White paper]. SANS Technology Institute. <https://sansorg.egnyte.com/dl/q7ICV3PndD>
- CrowdStrike Documentation. (2026). <https://falcon.crowdstrike.com/documentation/209/prevention-policy-settings>
NOTE: (Access to CrowdStrike Documentation requires an account/subscription)
- MITRE ATT&CK®. (n.d.). *ATT&CK®*. The MITRE Corporation. Retrieved February 28, 2026, from <https://attack.mitre.org/>

Atomic Red Team™. (n.d.). Retrieved February 23, 2026, from

<https://www.atomicredteam.io/>

CrowdStrike (n.d.). FALCON 102: Falcon Platform Onboarding Configuration.
CrowdStrike University.

<https://university.crowdstrike.com/learn/courses/472/falcon-102-falcon-platform-onboarding-configuration>

Arkaan, N. (2022, October 20). Medium. “CrowdStrike Falcon Series: Deployment to Maximum Protection” <https://medium.com/fmisec/crowdstrike-falcon-series-deployment-to-maximum-protection-5ba791d33270>

Sentonas, M. (2023, September 20). CrowdStrike Achieves 100% Protection, 100% Visibility, 100% Analytic Detection in MITRE Engenuity ATT&CK® Evaluations: Enterprise. CrowdStrike. <https://www.crowdstrike.com/en-us/blog/crowdstrike-achieves-100-percent-mitre-engenuity-attack-enterprise-evaluation/>

SOCFortress (2025, June 27). Medium. “How to Run Atomic Red Team on Linux and Automate Attack Simulations with Velociraptor” <https://socfortress.medium.com/how-to-run-atomic-red-team-on-linux-and-automate-attack-simulations-with-velociraptor-d4b52b05721b>

Thatipalli, S.A. (2023, November 10). Packt. “How to utilize Atomic Red Team Framework for Sophisticated Threat Simulations” <https://security.packt.com/how-to-utilize-atomic-red-team-framework-for-sophisticated-threat-simulations/>

Appendices

A.1 CrowdStrike Falcon Recommended Policy Settings

Setting Name	Phase 1 - initial deployment	Phase 2 - interim protection	Phase 3 - optimal protection
Sensor capabilities			
End user notifications	*	*	*
Sensor tamper prevention	Y	Y	Y
Sensor visibility Enhanced visibility			
Additional user mode data visibility	Y	Y	Y
Interpreter-only visibility	Y	Y	Y
System management engine visibility	Y	Y	Y
Script-based execution visibility		Y	Y
HTTP visibility and detection	Y	Y	Y
Redacted HTTP detection details	*	*	*
Hardware-enhanced exploit detection	Y	Y	Y
Enhanced exploitation visibility	Y	Y	Y
Enhanced DLL load visibility	-	-	-
WSL2 visibility	-	-	-
Sensor visibility Hardware-enhanced visibility			
Memory scanning with CPU		Y	Y
Sensor visibility Firmware			
BIOS firmware deep visibility			Y
Next-gen antivirus Sensor machine learning			
Enhanced machine learning for larger files		Y	Y
Next-gen antivirus On-demand scans			
USB insertion triggered scan	-	-	-
Next-gen antivirus On write			
Detect on write	Y	Y	Y
Quarantine on write		Y	Y
On write script file visibility	Y	Y	Y
Next-gen antivirus Quarantine			
Quarantine & security center registration		Y	Y
Quarantine on removable media		Y	Y
Next-gen antivirus Clean infected Microsoft Office files			
Microsoft Office file malicious macro removal			

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Malware protection Execution blocking			
Custom indicator blocking	Y	Y	Y
Suspicious process prevention		Y	Y
Suspicious registry operation prevention		Y	Y
Boot configuration database protection	-	-	-
Suspicious script and command prevention		Y	Y
Intelligence-sourced threat prevention	Y	Y	Y
Driver load prevention	Y	Y	Y
Vulnerable driver protection	Y	Y	Y
File system containment	-	-	-
Behavior-based prevention Exploit mitigation			
ASLR bypass prevention			Y
DEP bypass prevention			
Heap spray pre-allocation prevention			Y
NULL page allocation prevention		Y	Y
SEH overwrite prevention		Y	Y
Behavior-based prevention Ransomware			
Backup deletion prevention	Y	Y	Y
Cryptowall prevention	Y	Y	Y
File encryption prevention	Y	Y	Y
Locky prevention	Y	Y	Y
File system access prevention	Y	Y	Y
Volume shadow copy - audit		Y	Y
Volume shadow copy - protect			Y
Behavior-based prevention Exploitation behavior			
Application exploitation prevention	Y	Y	Y
Chopper webshell prevention	Y	Y	Y
Drive-by download prevention	Y	Y	Y
Code injection prevention	Y	Y	Y
JavaScript execution via Rundll32 prevention	Y	Y	Y
Behavior-based prevention Lateral movement and credential access			
Windows logon bypass ("Sticky keys") prevention	Y	Y	Y
Credential dumping prevention	Y	Y	Y
Behavior-based prevention Remediation			
Advanced remediation	Y	Y	Y
Sensor visibility Enhanced visibility			
Extended user mode data visibility			
Detection:	Disabled	Cautious	Moderate
Next-gen antivirus Microsoft Office file macro machine learning			

Cloud anti-malware for Microsoft Office files				
Detection:	Moderate	Aggressive	Aggressive	
Prevention:	Disabled	Moderate	Aggressive	
Next-gen antivirus Cloud machine learning				
Cloud-based anti-malware				
Detection:	Moderate	Aggressive	Aggressive	
Prevention:	Disabled	Moderate	Aggressive	
Cloud-based adware & pup				
Detection:	Moderate	Aggressive	Aggressive	
Prevention:	Disabled	Moderate	Aggressive	
Next-gen antivirus Sensor machine learning				
Sensor-based anti-malware				
Detection:	Moderate	Aggressive	Aggressive	
Prevention:	Disabled	Moderate	Aggressive	
Next-gen antivirus On-demand scans machine learning				
Cloud-based anti-malware on-demand scanning				
Detection:	-	-	-	
Prevention:	-	-	-	
Sensor-based anti-malware on-demand scanning				
Detection:	-	-	-	
Prevention:	-	-	-	
Cloud-based adware & pup on-demand scanning				
Detection:	-	-	-	
Prevention:	-	-	-	
Cloud-based detections Behavioral detections				
Cloud-based anomalous process execution				
Detection:	-	-	-	

*: Customer Preference

-: No recommendation

A.2 Policy Delta Results

The following tables list the unique fields that contained values under the listed policy setting, which did not contain values under the previous policy setting.

Additional User Mode Data Visibility
assessments.additional_user_mode_data_enabled
assessments.automated_remediation
assessments.beta_build_disabled
assessments.branch_target_injection_mitigation
assessments.branch_target_injection_mitigation_hardware_support
assessments.branch_target_injection_mitigation_patch
assessments.branch_target_injection_mitigation_registry_allowed
assessments.credential_guard_running
assessments.debug_mode_disabled
assessments.dma_guard_enabled
assessments.engine_full_visibility
assessments.execution_blocking_custom_blocking_enabled
assessments.execution_blocking_intel_threats_enabled
assessments.execution_blocking_suspicious_processes_enabled
assessments.execution_blocking_suspicious_registry_ops_enabled
assessments.execution_blocking_suspicious_scripts_enabled
assessments.exploit_mitigation_force_aslr_enabled
assessments.exploit_mitigation_heap_spray_allocation_enabled
assessments.exploit_mitigation_null_page_allocation_enabled
assessments.exploit_mitigation_seh_overwrite_protection_enabled
assessments.exploitation_behavior_application_exploitation_activity_enabled
assessments.exploitation_behavior_chopper_webshell_enabled
assessments.exploitation_behavior_code_injection_enabled
assessments.exploitation_behavior_driveby_download_enabled
assessments.exploitation_behavior_javascript_execution_rundll32_enabled
assessments.firmware_is_uefi
assessments.hardware_enhanced_exploit_detection
assessments.hsti_available
assessments.http_detections
assessments.hvci_enabled
assessments.hvci_strict_mode
assessments.in_full_functionality
assessments.interpreter_only

assessments.iommu_available
assessments.iommu_in_use
assessments.kmci_enabled
assessments.ll_terminal_fault_mitigation
assessments.lateral_movement_credential_access_credential_dumping_enabled
assessments.lateral_movement_credential_access_windows_logon_bypass_enabled
assessments.mbec_available
assessments.ml_adware
assessments.ml_adware_prevention
assessments.ml_cloud_antimalware
assessments.ml_cloud_antimalware_prevention
assessments.ml_sensor_antimalware
assessments.ml_sensor_antimalware_prevention
assessments.quarantine_and_security_registration
assessments.quarantine_on_write
assessments.ransomware_backup_deletion_enabled
assessments.ransomware_cryptowall_enabled
assessments.ransomware_file_encryption_enabled
assessments.ransomware_file_system_access_enabled
assessments.ransomware_locky_enabled
assessments.real_time_response_enabled
assessments.rogue_data_cache_load_mitigation
assessments.rogue_data_cache_load_mitigation_patch
assessments.script_based_execution_monitoring_enabled
assessments.script_enforcement
assessments.secure_boot_enabled
assessments.secure_kernel_running
assessments.secure_mor_available
assessments.sensor_tampering_protection
assessments.smm_protections
assessments.speculative_store_bypass_mitigation_available
assessments.speculative_store_bypass_mitigation_hardware_support
assessments.suspicious_kernel_drivers
assessments.system_firmware_bios_enabled
assessments.test_signing_disabled
assessments.uefi_memory_protection
assessments.volume_shadow_copy_audit
assessments.volume_shadow_copy_protect
assessments.vsm_available

assessments.windows_insider_program_disabled
assessments.windows_insider_program_not_running
assessments.windows_os_build
device_control_applied
device_control_applied_date
device_control_assigned_date
device_control_policy_id
event_type
group_id
group_type
host_hidden_status
os_version
prevention_applied
prevention_applied_date
prevention_assigned_date
prevention_policy_id
product_name
product_type_desc
remote_response_applied
rtr_state
scores.modified_time
scores.os
scores.overall
scores.sensor
scores.version
sensor_update_applied
sensor_update_applied_date
sensor_update_assigned_date
sensor_update_policy_id
sensor_update_policy_type

Interpreter-Only Visibility	
Attributes.action_target_name	
Attributes.actor_cid	
Attributes.success	
Attributes.target_cid	
Attributes.target_name	
Attributes.target_uuid	
Attributes.trace_id	

Script-based execution visibility & Quarantine & security center registration *	
AssociatedFile	
host.name	
IOCType	
IOCValue	

A.3 Atomic Red Team Testing

Tactic	Technique ID	Technique Name
command-and-control	T1001.002	Data Obfuscation via Steganography
credential-access	T1003	OS Credential Dumping
credential-access	T1003.001	OS Credential Dumping: LSASS Memory
credential-access	T1003.002	OS Credential Dumping: Security Account Manager
credential-access	T1003.003	OS Credential Dumping: NTDS
credential-access	T1003.004	OS Credential Dumping: LSA Secrets
credential-access	T1003.005	OS Credential Dumping: Cached Domain Credentials
credential-access	T1003.006	OS Credential Dumping: DCSync
collection	T1005	Data from Local System
defense-evasion	T1006	Direct Volume Access
discovery	T1007	System Service Discovery
discovery	T1010	Application Window Discovery
discovery	T1012	Query Registry
discovery	T1016	System Network Configuration Discovery
discovery	T1016.001	System Network Configuration Discovery: Internet Connection Discovery
discovery	T1016.002	System Network Configuration Discovery: Wi-Fi Discovery
discovery	T1018	Remote System Discovery
exfiltration	T1020	Automated Exfiltration
lateral-movement	T1021.001	Remote Services: Remote Desktop Protocol
lateral-movement	T1021.002	Remote Services: SMB/Windows Admin Shares
lateral-movement	T1021.003	Remote Services: Distributed Component Object Model
lateral-movement	T1021.006	Remote Services: Windows Remote Management
collection	T1025	Data from Removable Media
defense-evasion	T1027	Obfuscated Files or Information
defense-evasion	T1027.004	Obfuscated Files or Information: Compile After Delivery
defense-evasion	T1027.006	HTML Smuggling
defense-evasion	T1027.007	Obfuscated Files or Information: Dynamic API Resolution
defense-evasion	T1027.013	Obfuscated Files or Information: Encrypted/Encoded File
exfiltration	T1030	Data Transfer Size Limits
discovery	T1033	System Owner/User Discovery
defense-evasion	T1036	Masquerading
defense-evasion	T1036.003	Masquerading: Rename System Utilities
defense-evasion	T1036.004	Masquerading: Masquerade Task or Service
defense-evasion	T1036.005	Masquerading: Match Legitimate Name or Location
defense-evasion	T1036.007	Masquerading: Double File Extension

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privilege-escalation	T1037.001	Boot or Logon Initialization Scripts: Logon Script (Windows)
persistence	T1037.001	Boot or Logon Initialization Scripts: Logon Script (Windows)
collection	T1039	Data from Network Shared Drive
credential-access	T1040	Network Sniffing
discovery	T1040	Network Sniffing
exfiltration	T1041	Exfiltration Over C2 Channel
discovery	T1046	Network Service Discovery
execution	T1047	Windows Management Instrumentation
exfiltration	T1048	Exfiltration Over Alternative Protocol
exfiltration	T1048.002	Exfiltration Over Alternative Protocol - Exfiltration Over Asymmetric Encrypted Non-C2 Protocol
exfiltration	T1048.003	Exfiltration Over Alternative Protocol: Exfiltration Over Unencrypted/Obfuscated Non-C2 Protocol
discovery	T1049	System Network Connections Discovery
privilege-escalation	T1053.002	Scheduled Task/Job: At
execution	T1053.002	Scheduled Task/Job: At
persistence	T1053.002	Scheduled Task/Job: At
privilege-escalation	T1053.005	Scheduled Task/Job: Scheduled Task
execution	T1053.005	Scheduled Task/Job: Scheduled Task
persistence	T1053.005	Scheduled Task/Job: Scheduled Task
defense-evasion	T1055	Process Injection
privilege-escalation	T1055	Process Injection
defense-evasion	T1055.001	Process Injection: Dynamic-link Library Injection
privilege-escalation	T1055.001	Process Injection: Dynamic-link Library Injection
defense-evasion	T1055.002	Process Injection: Portable Executable Injection
privilege-escalation	T1055.002	Process Injection: Portable Executable Injection
defense-evasion	T1055.003	Thread Execution Hijacking
privilege-escalation	T1055.003	Thread Execution Hijacking
defense-evasion	T1055.004	Process Injection: Asynchronous Procedure Call
privilege-escalation	T1055.004	Process Injection: Asynchronous Procedure Call
defense-evasion	T1055.011	Process Injection: Extra Window Memory Injection
privilege-escalation	T1055.011	Process Injection: Extra Window Memory Injection
defense-evasion	T1055.012	Process Injection: Process Hollowing
privilege-escalation	T1055.012	Process Injection: Process Hollowing
defense-evasion	T1055.015	Process Injection: ListPlanting
privilege-escalation	T1055.015	Process Injection: ListPlanting
collection	T1056.001	Input Capture: Keylogging
credential-access	T1056.001	Input Capture: Keylogging
collection	T1056.002	Input Capture: GUI Input Capture
credential-access	T1056.002	Input Capture: GUI Input Capture

collection	T1056.004	Input Capture: Credential API Hooking
credential-access	T1056.004	Input Capture: Credential API Hooking
discovery	T1057	Process Discovery
execution	T1059	Command and Scripting Interpreter
execution	T1059.001	Command and Scripting Interpreter: PowerShell
execution	T1059.003	Command and Scripting Interpreter: Windows Command Shell
execution	T1059.005	Command and Scripting Interpreter: Visual Basic
execution	T1059.007	Command and Scripting Interpreter: JavaScript
execution	T1059.010	Command and Scripting Interpreter: AutoHotKey & AutoIT
discovery	T1069.001	Permission Groups Discovery: Local Groups
discovery	T1069.002	Permission Groups Discovery: Domain Groups
defense-evasion	T1070	Indicator Removal on Host
defense-evasion	T1070.001	Indicator Removal on Host: Clear Windows Event Logs
defense-evasion	T1070.003	Indicator Removal on Host: Clear Command History
defense-evasion	T1070.004	Indicator Removal on Host: File Deletion
defense-evasion	T1070.005	Indicator Removal on Host: Network Share Connection Removal
defense-evasion	T1070.006	Indicator Removal on Host: Timestamp
defense-evasion	T1070.008	Email Collection: Mailbox Manipulation
command-and-control	T1071	Application Layer Protocol
command-and-control	T1071.001	Application Layer Protocol: Web Protocols
command-and-control	T1071.004	Application Layer Protocol: DNS
execution	T1072	Software Deployment Tools
lateral-movement	T1072	Software Deployment Tools
collection	T1074.001	Data Staged: Local Data Staging
defense-evasion	T1078.001	Valid Accounts: Default Accounts
privilege-escalation	T1078.001	Valid Accounts: Default Accounts
persistence	T1078.001	Valid Accounts: Default Accounts
initial-access	T1078.001	Valid Accounts: Default Accounts
defense-evasion	T1078.003	Valid Accounts: Local Accounts
privilege-escalation	T1078.003	Valid Accounts: Local Accounts
persistence	T1078.003	Valid Accounts: Local Accounts
initial-access	T1078.003	Valid Accounts: Local Accounts
discovery	T1082	System Information Discovery
discovery	T1083	File and Directory Discovery
discovery	T1087.001	Account Discovery: Local Account
discovery	T1087.002	Account Discovery: Domain Account

command-and-control	T1090.001	Proxy: Internal Proxy
command-and-control	T1090.003	Proxy: Multi-hop Proxy
lateral-movement	T1091	Replication Through Removable Media
initial-access	T1091	Replication Through Removable Media
command-and-control	T1095	Non-Application Layer Protocol
privilege-escalation	T1098	Account Manipulation
persistence	T1098	Account Manipulation
command-and-control	T1105	Ingress Tool Transfer
execution	T1106	Native API
credential-access	T1110.001	Brute Force: Password Guessing
credential-access	T1110.002	Brute Force: Password Cracking
credential-access	T1110.003	Brute Force: Password Spraying
credential-access	T1110.004	Brute Force: Credential Stuffing
defense-evasion	T1112	Modify Registry
persistence	T1112	Modify Registry
collection	T1113	Screen Capture
collection	T1114.001	Email Collection: Local Email Collection
collection	T1115	Clipboard Data
collection	T1119	Automated Collection
discovery	T1120	Peripheral Device Discovery
collection	T1123	Audio Capture
discovery	T1124	System Time Discovery
collection	T1125	Video Capture
defense-evasion	T1127	Trusted Developer Utilities Proxy Execution
defense-evasion	T1127.001	Trusted Developer Utilities Proxy Execution: MSBuild
execution	T1129	Server Software Component
command-and-control	T1132.001	Data Encoding: Standard Encoding
persistence	T1133	External Remote Services
initial-access	T1133	External Remote Services
defense-evasion	T1134.001	Access Token Manipulation: Token Impersonation/Theft
privilege-escalation	T1134.001	Access Token Manipulation: Token Impersonation/Theft
defense-evasion	T1134.002	Create Process with Token
privilege-escalation	T1134.002	Create Process with Token
defense-evasion	T1134.004	Access Token Manipulation: Parent PID Spoofing
privilege-escalation	T1134.004	Access Token Manipulation: Parent PID Spoofing
defense-evasion	T1134.005	Access Token Manipulation: SID-History Injection

privilege-escalation	T1134.005	Access Token Manipulation: SID-History Injection
discovery	T1135	Network Share Discovery
persistence	T1136.001	Create Account: Local Account
persistence	T1136.002	Create Account: Domain Account
persistence	T1137	Office Application Startup
persistence	T1137.001	Office Application Startup: Office Template Macros.
persistence	T1137.002	Office Application Startup: Office Test
persistence	T1137.004	Office Application Startup: Outlook Home Page
persistence	T1137.006	Office Application Startup: Add-ins
defense-evasion	T1140	Deobfuscate/Decode Files or Information
persistence	T1176	Browser Extensions
credential-access	T1187	Forced Authentication
initial-access	T1195	Supply Chain Compromise
defense-evasion	T1197	BITS Jobs
persistence	T1197	BITS Jobs
discovery	T1201	Password Policy Discovery
defense-evasion	T1202	Indirect Command Execution
execution	T1204.002	User Execution: Malicious File
defense-evasion	T1207	Rogue Domain Controller
defense-evasion	T1216	Signed Script Proxy Execution
defense-evasion	T1216.001	Signed Script Proxy Execution: Pubprn
discovery	T1217	Browser Bookmark Discovery
defense-evasion	T1218	Signed Binary Proxy Execution
defense-evasion	T1218.001	Signed Binary Proxy Execution: Compiled HTML File
defense-evasion	T1218.002	Signed Binary Proxy Execution: Control Panel
defense-evasion	T1218.003	Signed Binary Proxy Execution: CMSTP
defense-evasion	T1218.004	Signed Binary Proxy Execution: InstallUtil
defense-evasion	T1218.005	Signed Binary Proxy Execution: Mshta
defense-evasion	T1218.007	Signed Binary Proxy Execution: Msiexec
defense-evasion	T1218.008	Signed Binary Proxy Execution: Odbcconf
defense-evasion	T1218.009	Signed Binary Proxy Execution: Regsvcs/Regasm
defense-evasion	T1218.010	Signed Binary Proxy Execution: Regsvr32
defense-evasion	T1218.011	Signed Binary Proxy Execution: Rundll32
command-and-control	T1219	Remote Access Software
defense-evasion	T1220	XSL Script Processing
defense-evasion	T1221	Template Injection
defense-evasion	T1222	File and Directory Permissions Modification
defense-evasion	T1222.001	File and Directory Permissions Modification: Windows File and Directory Permissions Modification

discovery	T1482	Domain Trust Discovery
defense-evasion	T1484.001	Domain Policy Modification: Group Policy Modification
privilege-escalation	T1484.001	Domain Policy Modification: Group Policy Modification
impact	T1485	Data Destruction
impact	T1486	Data Encrypted for Impact
impact	T1489	Service Stop
impact	T1490	Inhibit System Recovery
impact	T1491.001	Defacement: Internal Defacement
impact	T1496	Resource Hijacking
defense-evasion	T1497.001	Virtualization/Sandbox Evasion: System Checks
discovery	T1497.001	Virtualization/Sandbox Evasion: System Checks
persistence	T1505.002	Server Software Component: Transport Agent
persistence	T1505.003	Server Software Component: Web Shell
persistence	T1505.004	IIS Components
persistence	T1505.005	Server Software Component: Terminal Services DLL
discovery	T1518	Software Discovery
discovery	T1518.001	Software Discovery: Security Software Discovery
impact	T1529	System Shutdown/Reboot
impact	T1531	Account Access Removal
credential-access	T1539	Steal Web Session Cookie
defense-evasion	T1542.001	Pre-OS Boot: System Firmware
persistence	T1542.001	Pre-OS Boot: System Firmware
privilege-escalation	T1543.003	Create or Modify System Process: Windows Service
persistence	T1543.003	Create or Modify System Process: Windows Service
privilege-escalation	T1546	Event Triggered Execution
persistence	T1546	Event Triggered Execution
privilege-escalation	T1546.001	Event Triggered Execution: Change Default File Association
persistence	T1546.001	Event Triggered Execution: Change Default File Association
privilege-escalation	T1546.002	Event Triggered Execution: Screensaver
persistence	T1546.002	Event Triggered Execution: Screensaver
privilege-escalation	T1546.003	Event Triggered Execution: Windows Management Instrumentation Event Subscription
persistence	T1546.003	Event Triggered Execution: Windows Management Instrumentation Event Subscription
privilege-escalation	T1546.007	Event Triggered Execution: Netsh Helper DLL
persistence	T1546.007	Event Triggered Execution: Netsh Helper DLL
privilege-escalation	T1546.008	Event Triggered Execution: Accessibility Features
persistence	T1546.008	Event Triggered Execution: Accessibility Features
privilege-escalation	T1546.009	Event Triggered Execution: AppCert DLLs
persistence	T1546.009	Event Triggered Execution: AppCert DLLs

privilege-escalation	T1546.010	Event Triggered Execution: Applnit DLLs
persistence	T1546.010	Event Triggered Execution: Applnit DLLs
privilege-escalation	T1546.011	Event Triggered Execution: Application Shimming
persistence	T1546.011	Event Triggered Execution: Application Shimming
privilege-escalation	T1546.012	Event Triggered Execution: Image File Execution Options Injection
persistence	T1546.012	Event Triggered Execution: Image File Execution Options Injection
privilege-escalation	T1546.013	Event Triggered Execution: PowerShell Profile
persistence	T1546.013	Event Triggered Execution: PowerShell Profile
privilege-escalation	T1546.015	Event Triggered Execution: Component Object Model Hijacking
persistence	T1546.015	Event Triggered Execution: Component Object Model Hijacking
privilege-escalation	T1546.018	Event Triggered Execution: Python Startup Hooks
persistence	T1546.018	Event Triggered Execution: Python Startup Hooks
privilege-escalation	T1547	Boot or Logon Autostart Execution
persistence	T1547	Boot or Logon Autostart Execution
privilege-escalation	T1547.001	Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder
persistence	T1547.001	Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder
privilege-escalation	T1547.002	Authentication Package
persistence	T1547.002	Authentication Package
privilege-escalation	T1547.003	Time Providers
persistence	T1547.003	Time Providers
privilege-escalation	T1547.004	Boot or Logon Autostart Execution: Winlogon Helper DLL
persistence	T1547.004	Boot or Logon Autostart Execution: Winlogon Helper DLL
privilege-escalation	T1547.005	Boot or Logon Autostart Execution: Security Support Provider
persistence	T1547.005	Boot or Logon Autostart Execution: Security Support Provider
privilege-escalation	T1547.008	Boot or Logon Autostart Execution: LSASS Driver
persistence	T1547.008	Boot or Logon Autostart Execution: LSASS Driver
privilege-escalation	T1547.009	Boot or Logon Autostart Execution: Shortcut Modification
persistence	T1547.009	Boot or Logon Autostart Execution: Shortcut Modification
privilege-escalation	T1547.010	Boot or Logon Autostart Execution: Port Monitors
persistence	T1547.010	Boot or Logon Autostart Execution: Port Monitors
privilege-escalation	T1547.012	Boot or Logon Autostart Execution: Print Processors
persistence	T1547.012	Boot or Logon Autostart Execution: Print Processors
privilege-escalation	T1547.014	Active Setup
persistence	T1547.014	Active Setup
defense-evasion	T1548.002	Abuse Elevation Control Mechanism: Bypass User Account Control
privilege-escalation	T1548.002	Abuse Elevation Control Mechanism: Bypass User Account Control
defense-evasion	T1550.002	Use Alternate Authentication Material: Pass the Hash
lateral-movement	T1550.002	Use Alternate Authentication Material: Pass the Hash
defense-evasion	T1550.003	Use Alternate Authentication Material: Pass the Ticket

lateral-movement	T1550.003	Use Alternate Authentication Material: Pass the Ticket
credential-access	T1552	Unsecured Credentials
credential-access	T1552.001	Unsecured Credentials: Credentials In Files
credential-access	T1552.002	Unsecured Credentials: Credentials in Registry
credential-access	T1552.004	Unsecured Credentials: Private Keys
credential-access	T1552.006	Unsecured Credentials: Group Policy Preferences
defense-evasion	T1553.003	Subvert Trust Controls: SIP and Trust Provider Hijacking
defense-evasion	T1553.004	Subvert Trust Controls: Install Root Certificate
defense-evasion	T1553.005	Subvert Trust Controls: Mark-of-the-Web Bypass
defense-evasion	T1553.006	Subvert Trust Controls: Code Signing Policy Modification
credential-access	T1555	Credentials from Password Stores
credential-access	T1555.003	Credentials from Password Stores: Credentials from Web Browsers
credential-access	T1555.004	Credentials from Password Stores: Windows Credential Manager
defense-evasion	T1556.002	Modify Authentication Process: Password Filter DLL
persistence	T1556.002	Modify Authentication Process: Password Filter DLL
credential-access	T1556.002	Modify Authentication Process: Password Filter DLL
collection	T1557.001	Adversary-in-the-Middle: LLMNR/NBT-NS Poisoning and SMB Relay
credential-access	T1557.001	Adversary-in-the-Middle: LLMNR/NBT-NS Poisoning and SMB Relay
credential-access	T1558.001	Steal or Forge Kerberos Tickets: Golden Ticket
credential-access	T1558.002	Steal or Forge Kerberos Tickets: Silver Ticket
credential-access	T1558.003	Steal or Forge Kerberos Tickets: Kerberoasting
credential-access	T1558.004	Steal or Forge Kerberos Tickets: AS-REP Roasting
execution	T1559	Inter-Process Communication
execution	T1559.002	Inter-Process Communication: Dynamic Data Exchange
collection	T1560	Archive Collected Data
collection	T1560.001	Archive Collected Data: Archive via Utility
defense-evasion	T1562	Impair Defenses
defense-evasion	T1562.001	Impair Defenses: Disable or Modify Tools
defense-evasion	T1562.002	Impair Defenses: Disable Windows Event Logging
defense-evasion	T1562.003	Impair Defenses: Impair Command History Logging
defense-evasion	T1562.004	Impair Defenses: Disable or Modify System Firewall
defense-evasion	T1562.006	Impair Defenses: Indicator Blocking
defense-evasion	T1562.009	Impair Defenses: Safe Boot Mode
defense-evasion	T1562.010	Impair Defenses: Downgrade Attack
lateral-movement	T1563.002	Remote Service Session Hijacking: RDP Hijacking
defense-evasion	T1564	Hide Artifacts
defense-evasion	T1564.001	Hide Artifacts: Hidden Files and Directories
defense-evasion	T1564.002	Hide Artifacts: Hidden Users
defense-evasion	T1564.003	Hide Artifacts: Hidden Window

defense-evasion	T1564.004	Hide Artifacts: NTFS File Attributes
defense-evasion	T1564.006	Run Virtual Instance
initial-access	T1566.001	Phishing: Spearphishing Attachment
initial-access	T1566.002	Phishing: Spearphishing Link
exfiltration	T1567.002	Exfiltration Over Web Service: Exfiltration to Cloud Storage
exfiltration	T1567.003	Exfiltration Over Web Service: Exfiltration to Text Storage Sites
execution	T1569.002	System Services: Service Execution
lateral-movement	T1570	Lateral Tool Transfer
command-and-control	T1571	Non-Standard Port
command-and-control	T1572	Protocol Tunneling
command-and-control	T1573	Encrypted Channel
defense-evasion	T1574.001	Hijack Execution Flow: DLL
privilege-escalation	T1574.001	Hijack Execution Flow: DLL
persistence	T1574.001	Hijack Execution Flow: DLL
defense-evasion	T1574.008	Hijack Execution Flow: Path Interception by Search Order Hijacking
privilege-escalation	T1574.008	Hijack Execution Flow: Path Interception by Search Order Hijacking
persistence	T1574.008	Hijack Execution Flow: Path Interception by Search Order Hijacking
defense-evasion	T1574.009	Hijack Execution Flow: Path Interception by Unquoted Path
privilege-escalation	T1574.009	Hijack Execution Flow: Path Interception by Unquoted Path
persistence	T1574.009	Hijack Execution Flow: Path Interception by Unquoted Path
defense-evasion	T1574.011	Hijack Execution Flow: Services Registry Permissions Weakness
privilege-escalation	T1574.011	Hijack Execution Flow: Services Registry Permissions Weakness
persistence	T1574.011	Hijack Execution Flow: Services Registry Permissions Weakness
defense-evasion	T1574.012	Hijack Execution Flow: COR_PROFILER
privilege-escalation	T1574.012	Hijack Execution Flow: COR_PROFILER
persistence	T1574.012	Hijack Execution Flow: COR_PROFILER
discovery	T1614	System Location Discovery
discovery	T1614.001	System Location Discovery: System Language Discovery
discovery	T1615	Group Policy Discovery
defense-evasion	T1620	Reflective Code Loading
defense-evasion	T1622	Debugger Evasion
discovery	T1622	Debugger Evasion
credential-access	T1649	Steal or Forge Authentication Certificates
discovery	T1652	Device Driver Discovery
discovery	T1654	Log Enumeration